



CIRCUIT THEORY (EE-217)

WIRELESS POWER TRANSFER CIRCUIT

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OBJECTIVE

The objective of this experiment is to design, construct, and test a wireless power transfer circuit that can transmit electrical energy from one coil to another without the use of physical wires. The aim is to understand the concept of electromagnetic induction and demonstrate how electrical energy can be transferred through a magnetic field over a short distance to power a load such as an LED.

APPARATUS / COMPONENTS

- BD139 NPN Transistor
- 10 k Ω Resistor
- 1 k Ω Resistor
- 1N4148 Diode
- 10 nF Ceramic Capacitor
- 100 μ F Electrolytic Capacitor
- LED (as load)
- Two Copper Coils (each 27 mm diameter, ~30 turns)
- 9 V DC Power Supply
- Connecting Wires
- Breadboard

INTRODUCTION

Wireless power transfer (WPT) is a modern technology that allows electrical energy to be transmitted from a power source to a device without physical conductors. It relies on the principles of **electromagnetic induction** or **resonant coupling**. In this project, a transmitter circuit generates a rapidly changing magnetic field using a coil and transistor-based oscillator. A nearby receiver coil captures this energy, converts it back into electric current, and powers a connected load.

This type of circuit demonstrates the basic working principle used in wireless charging systems, such as those found in electric toothbrushes and mobile phones. It is a simple yet effective model for understanding the fundamentals of contactless energy transfer.

THEORY

The wireless power transfer circuit operates on the principle of **Faraday's Law of Electromagnetic Induction**, which states that a changing magnetic field induces an electromotive force (EMF) in a nearby conductor.

In this circuit, the **transmitter coil** is connected to a transistor oscillator circuit powered by a DC supply. The transistor acts as a high-speed switch, continuously turning on and off, causing current to alternate in the coil. This alternating current produces a changing magnetic field around the transmitter coil.

When the **receiver coil** is placed within this magnetic field, a voltage is induced across it. The induced current flows through a diode and capacitor to convert it into DC, which then lights up the LED. The brightness of the LED depends on the strength of the magnetic coupling between the two coils and the distance separating them.

The efficiency of power transfer decreases as the distance between coils increases because the magnetic field weakens with distance. Alignment of coils also plays a significant role in maintaining strong coupling and efficient power transfer.

CIRCUIT DESCRIPTION

The circuit is divided into two main sections:

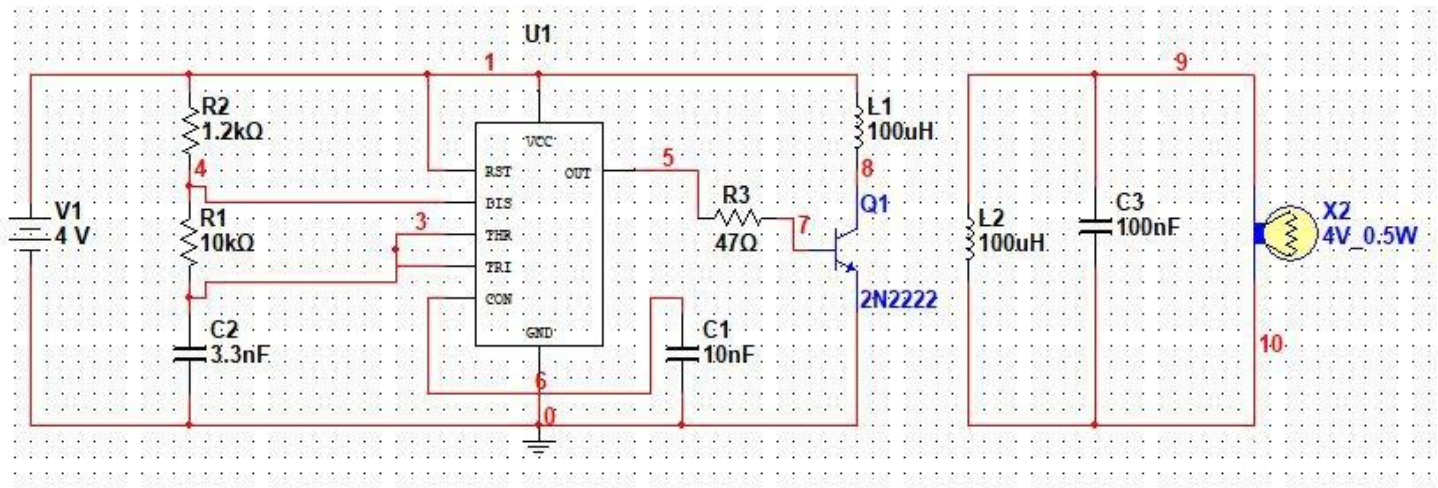
1. **Transmitter Section**

- The transmitter uses a BD139 transistor along with resistors and a diode.
- When powered by a 9 V DC supply, the transistor rapidly switches current through the transmitter coil.
- This generates an alternating magnetic field around the coil.

- The circuit is self-oscillating and requires no external frequency source.

2. Receiver Section

- The receiver consists of a coil, diode, capacitor, and LED.
- The coil picks up energy from the changing magnetic field of the transmitter.
- The diode rectifies the induced AC current, and the capacitor smooths it into DC to power the LED.



PROCEDURE

1. Two identical coils were made using enameled copper wire, each consisting of approximately 30 turns wound around a 27 mm diameter circular form.
2. The transmitter circuit was built on a breadboard using a BD139 transistor, two resistors, one diode, and the transmitter coil.
3. The receiver circuit was made separately with a similar coil, a diode, a capacitor, and an LED connected as a load.
4. The transmitter circuit was connected to a 9 V DC power source.
5. The receiver coil was placed near the transmitter coil, and the LED was observed for illumination.

6. The distance between coils was gradually increased to study the effect on LED brightness.
7. Observations were recorded for performance, stability, and distance range.

APPLICATIONS

1. Wireless charging for smartphones and wearable devices
2. Electric toothbrushes and shavers
3. Charging pads for electric vehicles (EVs)
4. Biomedical implants requiring non-contact power
5. Industrial applications for rotating or sealed equipment

OBSERVATIONS

Parameter	Observation
Power Supply	9 V DC
Coil Diameter	27 mm
Number of Turns	~30 turns each
Type of Wire	Enameled Copper Wire
Load	LED
Maximum Transfer Distance Around 2–3 cm	
Coil Alignment	Coaxial alignment gives best result
LED Brightness	High when coils are close, dim when moved apart

RESULT

The experiment successfully demonstrated the wireless transfer of electrical energy using electromagnetic induction. The LED in the receiver circuit glowed when the coils were placed close to each other, confirming that energy was transmitted wirelessly through the magnetic field. The system worked effectively for short distances and showed that coil alignment and distance have a major impact on power transfer efficiency.

CONCLUSION

From this experiment, it can be concluded that wireless power transfer through inductive coupling is possible using simple electronic components. The BD139 transistor efficiently created oscillations necessary for generating the alternating magnetic field in the transmitter coil.

The setup proved that as the distance between the coils increases, the coupling decreases and less energy is transferred. This project provides a basic understanding of how modern wireless charging devices function and how electromagnetic principles can be applied in practical circuits.

Overall, the circuit worked reliably, demonstrating clear wireless transmission of power and offering insights into improving range and efficiency in future designs.